

Polynomial Regression



בין שני הקצוות האלה
יש אין-סוף אפשרויות באמצע.



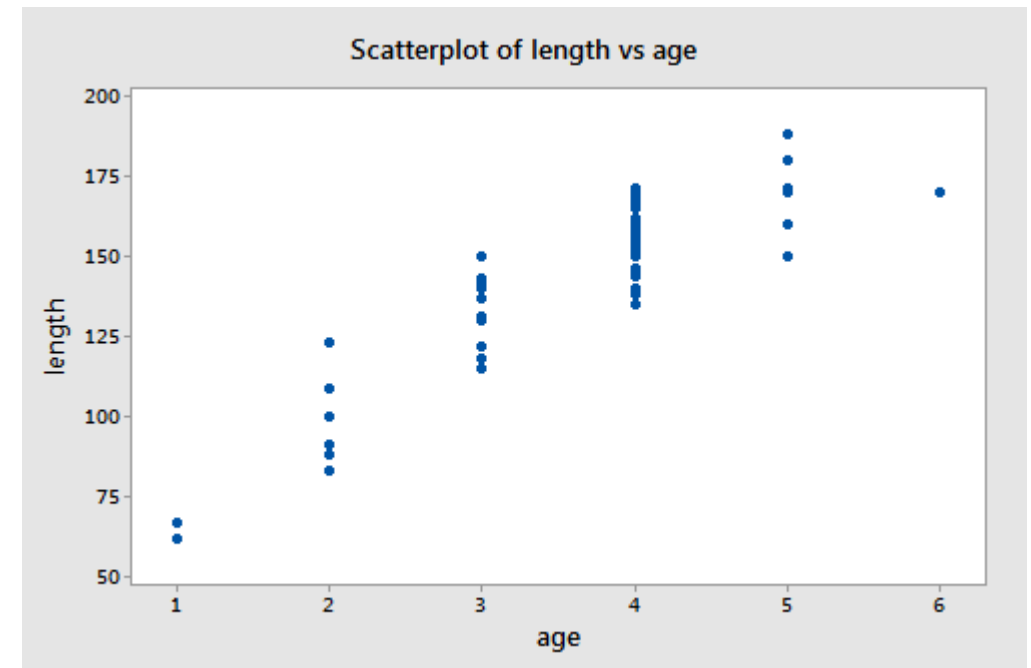
age	length
1	67
1	62
2	109
2	83
2	91
2	88
3	137
3	131
3	122
3	122
3	118
3	115
3	131
3	143
3	142
2	123
3	122
4	138
4	135
4	146
4	146
4	145

Polynomial regression - Example

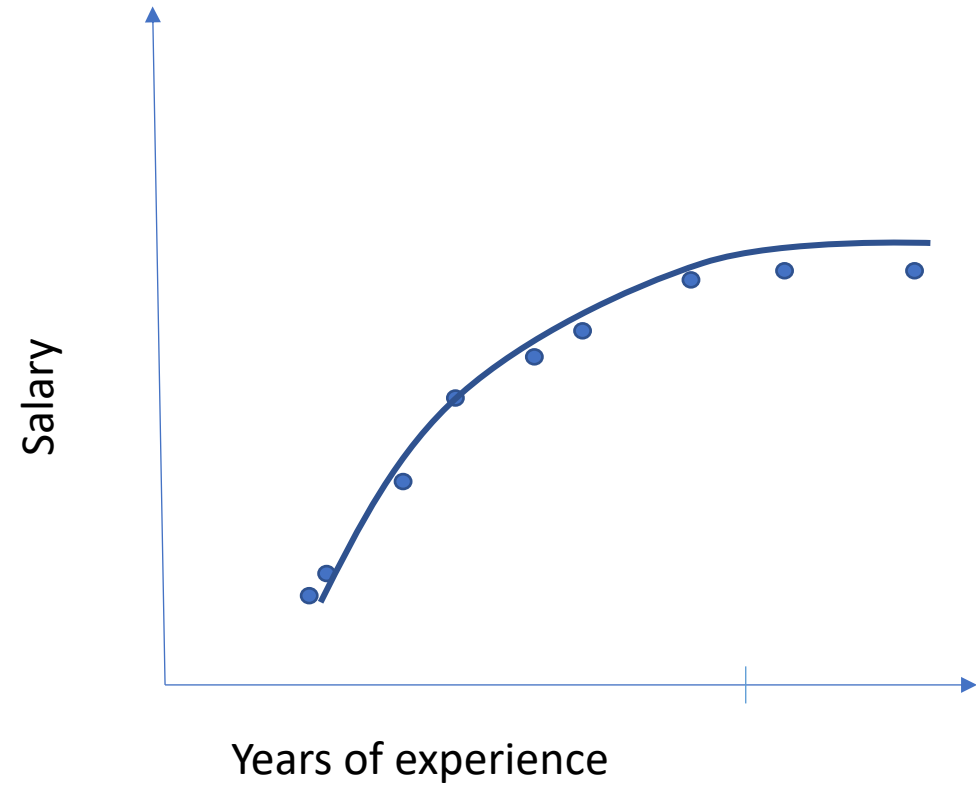
In 1981, 78 bluegills were randomly sampled from Lake Mary in Minnesota. The researchers (Cook and Weisberg, 1999) measured and recorded the length (in mm) of the fish and the age (in years) of the fish.



By Ltshears - Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=5856538>



Linear Regression VS. Polynomial Regression



Polynomial Regression

- Simple Linear Regression: $y = b_0 + b_1x_1$
- Multiple Linear Regression: $y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$
- Polynomial Regression: $y = b_0 + b_1x_1 + b_2x_1^2 + \cdots + b_nx_1^n$

Polynomial Regression - Example

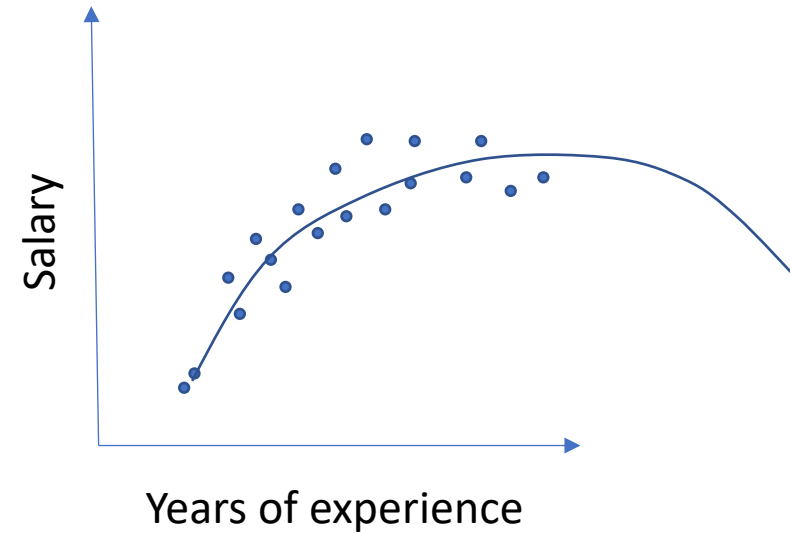


Quadratic model

$$y = b_0 + b_1x_1 + b_2x_1^2$$



Polynomial Regression - Example



Quadratic model $y = b_0 + b_1x_1 + b_2 x_1^2$

Polynomial Regression - Example



Cubic model

$$y = b_0 + b_1x_1 + b_2x_1^2 + b_3x_1^3$$



Polynomial Regression & Multiple Linear Regression

Multiple Linear Regression: $y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$

Polynomial Regression: $y = b_0 + b_1x_1 + b_2x_1^2 + \cdots + b_nx_1^n$

$$y = b_0 + b_1e_1 + b_2e_2 + \cdots + b_ne_n$$

$$\begin{aligned}e_1 &= x_1 \\e_2 &= x_1^2 \\e_n &= x_1^n\end{aligned}$$

Example

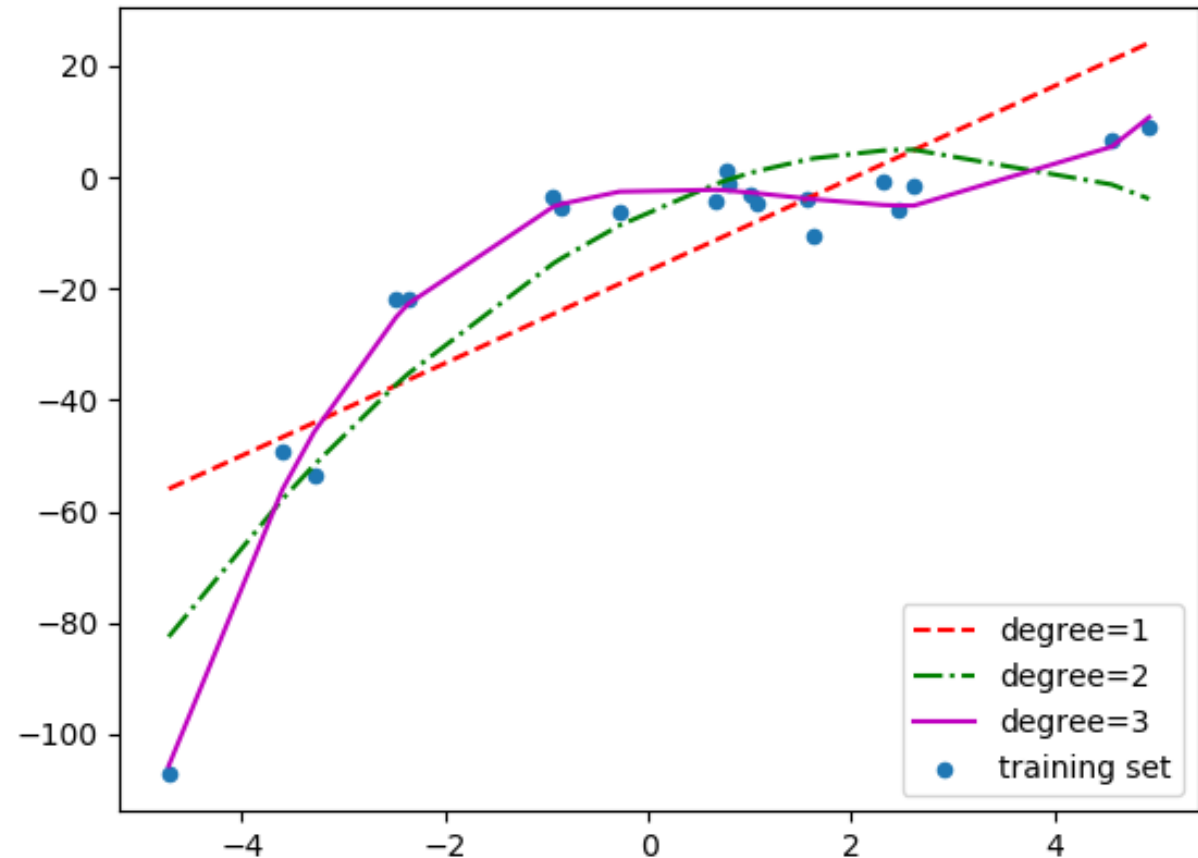
- Red line – linear regression
- Green line – $ax^2 + bx + c$
- Purple line – $ax^3 + bx^2 + cx + d$

Which is the best?

Red:
RMSE of linear regression is **15.90**
 R^2 score of linear regression is **0.63**

Green:
RMSE of polynomial regression is **10.12**
 R^2 of polynomial regression is **0.85**

Purple:
RMSE is **3.44**
 R^2 score is **0.98**

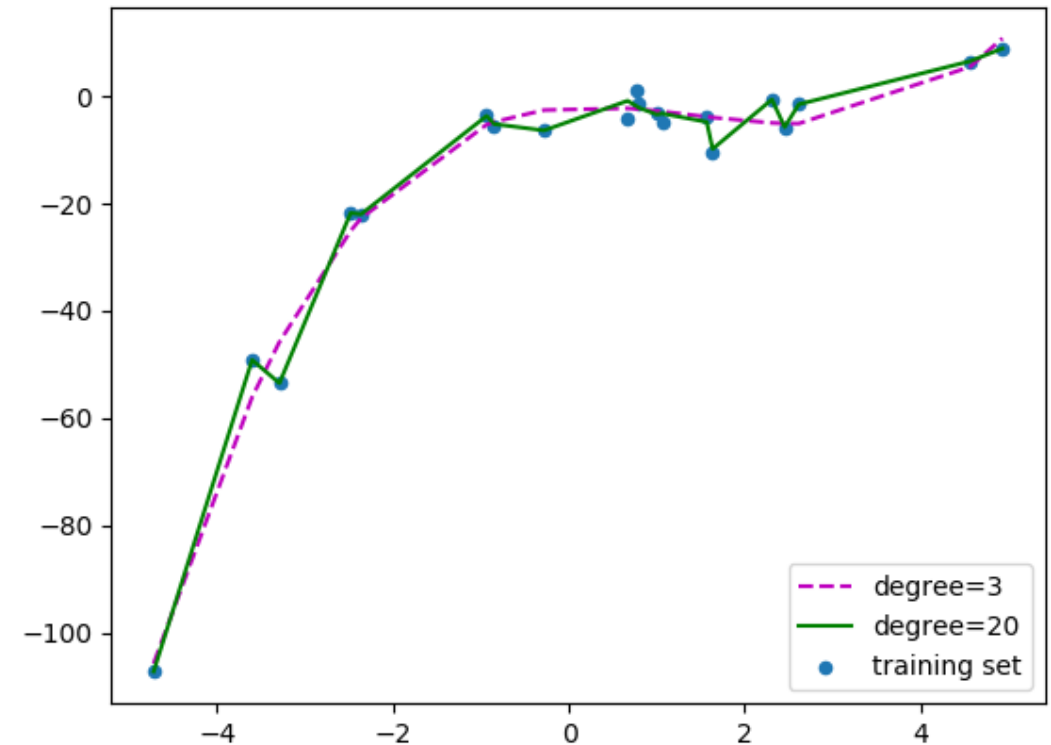


Even higher degree...

$$\begin{aligned} & x^{10} + 4ax^9 + (6a^2 + 3b)x^8 + (4a^3 + 9ab + c)x^7 + \\ & (a^4 + 9a^2b + 3b^2 + 4ac - 3d)x^6 + (3a^3b + 6ab^2 + 5a^2c + 2bc - 5ad - 11e)x^5 + \\ & (3a^2b^2 + b^3 + 2a^3c + 6abc - c^2 - 2a^2d - 2bd - 22ae)x^4 + \\ & (ab^3 + 4a^2bc + b^2c - 4cd - 16a^2e - 4be)x^3 + \\ & (2ab^2c + a^2c^2 - bc^2 + a^2bd + b^2d - 3acd - 4d^2 - 4a^3e - 9abe + 7ce)x^2 + \\ & (abc^2 - c^3 + ab^2d - 4ad^2 - 4a^2be - b^2e + 4ace + 4de)x + \\ & (abcd - c^2d - a^2d^2 - ab^2e + bce + 2ade - e^2). \end{aligned}$$

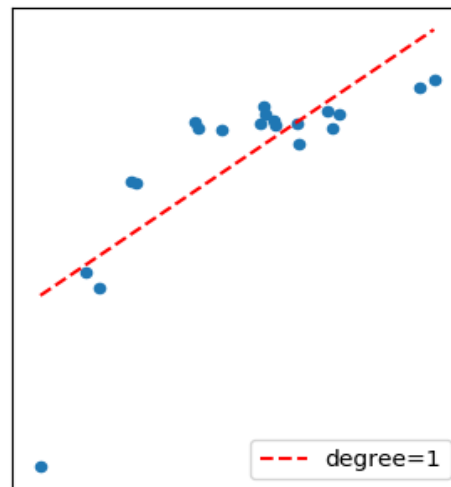
Degree = 20

- Problem: **overfitting**
- The model captures the noises in the data
- **Solution:** add more training points
 - Not always feasible
- **How to choose the optimal model?**

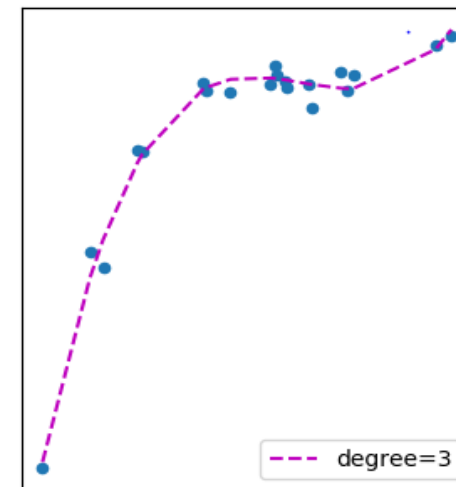


The Bias vs Variance trade-off

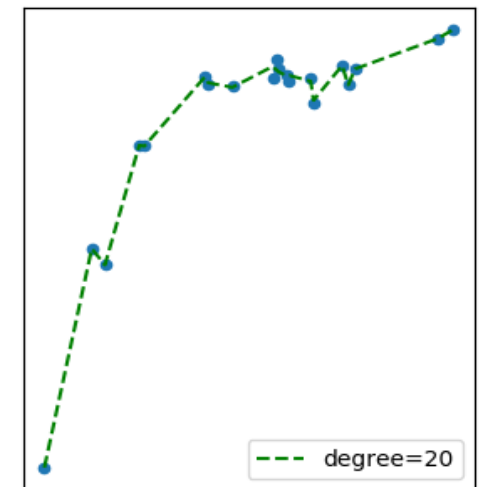
- **Bias:** the error due to the model's simplistic assumptions in fitting the data. A high bias means that the model is unable to capture the patterns in the data and this results in **under-fitting**.
- **Variance:** the error due to the complex model trying to fit the data. High variance means the model passes through most of the data points and it results in **over-fitting** the data.



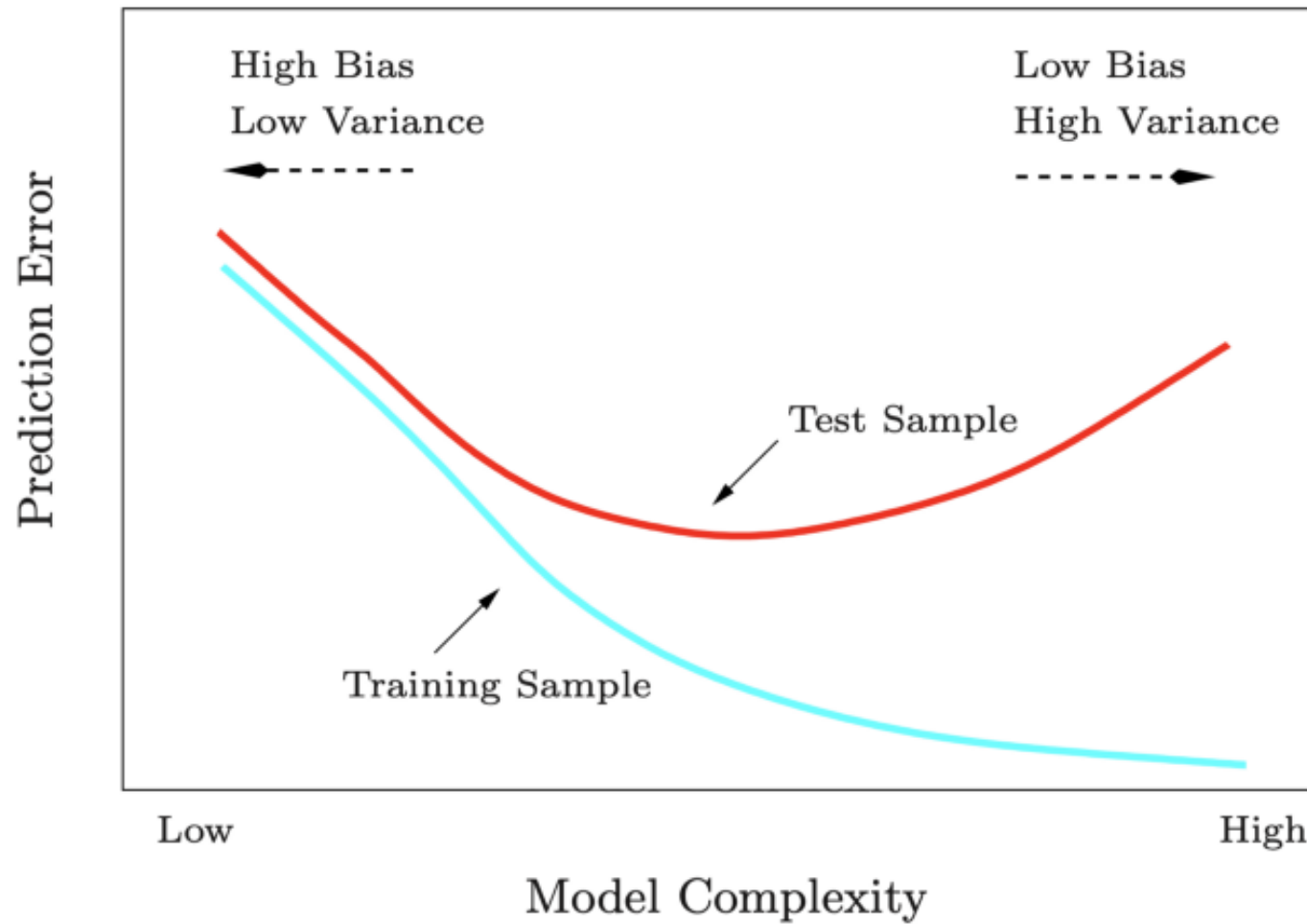
Underfit
High Bias
Low Variance



Correct Fit
Low Bias
Low Variance

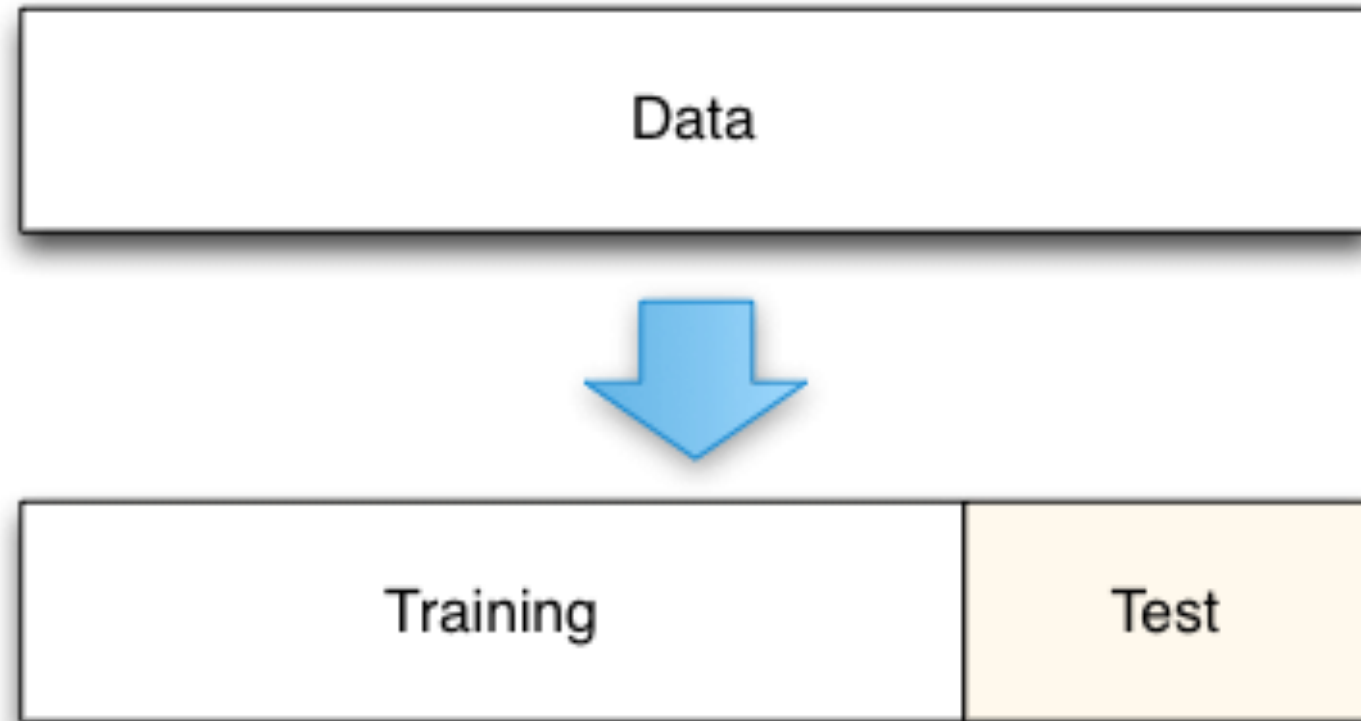


Overfit
Low Bias
High Variance



Bias Variance
trade-off

Train-Test Data Split





Validation Techniques

Resubstitution

Hold-out

K-fold cross-validation

LOOCV

Random subsampling

Bootstrapping

Resubstitution



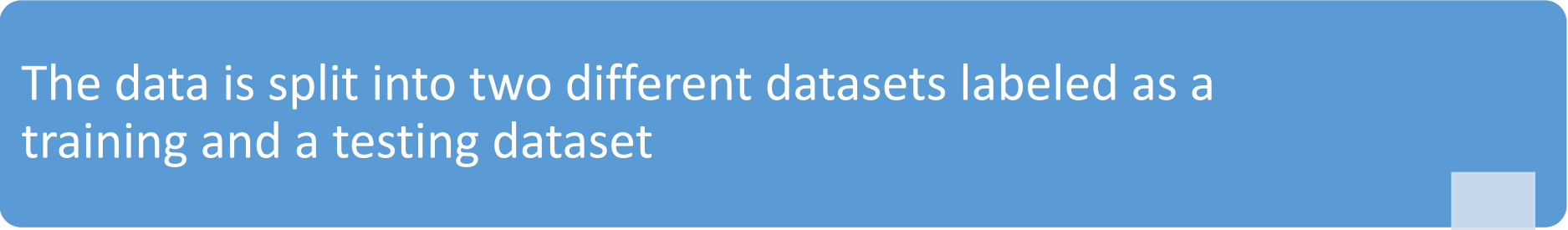
All the data is used for training the model



The error rate is evaluated from the same training data set

Holdout

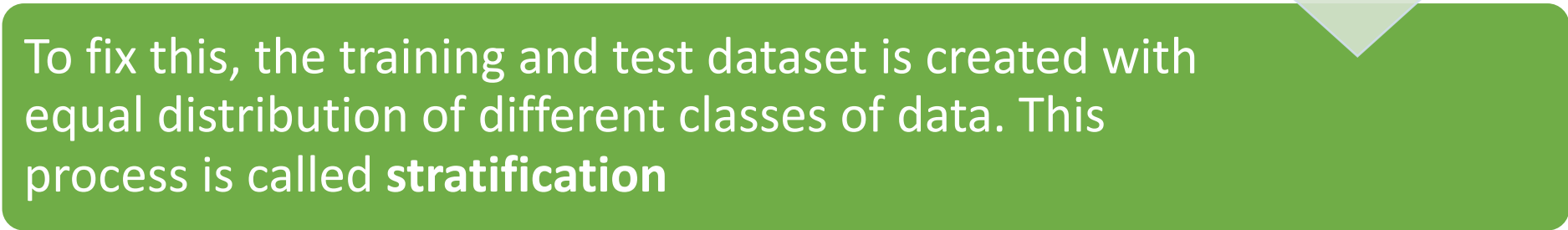
The data is split into two different datasets labeled as a training and a testing dataset



There is a likelihood that uneven distribution of different classes of data is found in training and test dataset



To fix this, the training and test dataset is created with equal distribution of different classes of data. This process is called **stratification**



K-Fold Cross-Validation

In this technique, $k-1$ folds are used for training and the remaining one is used for testing

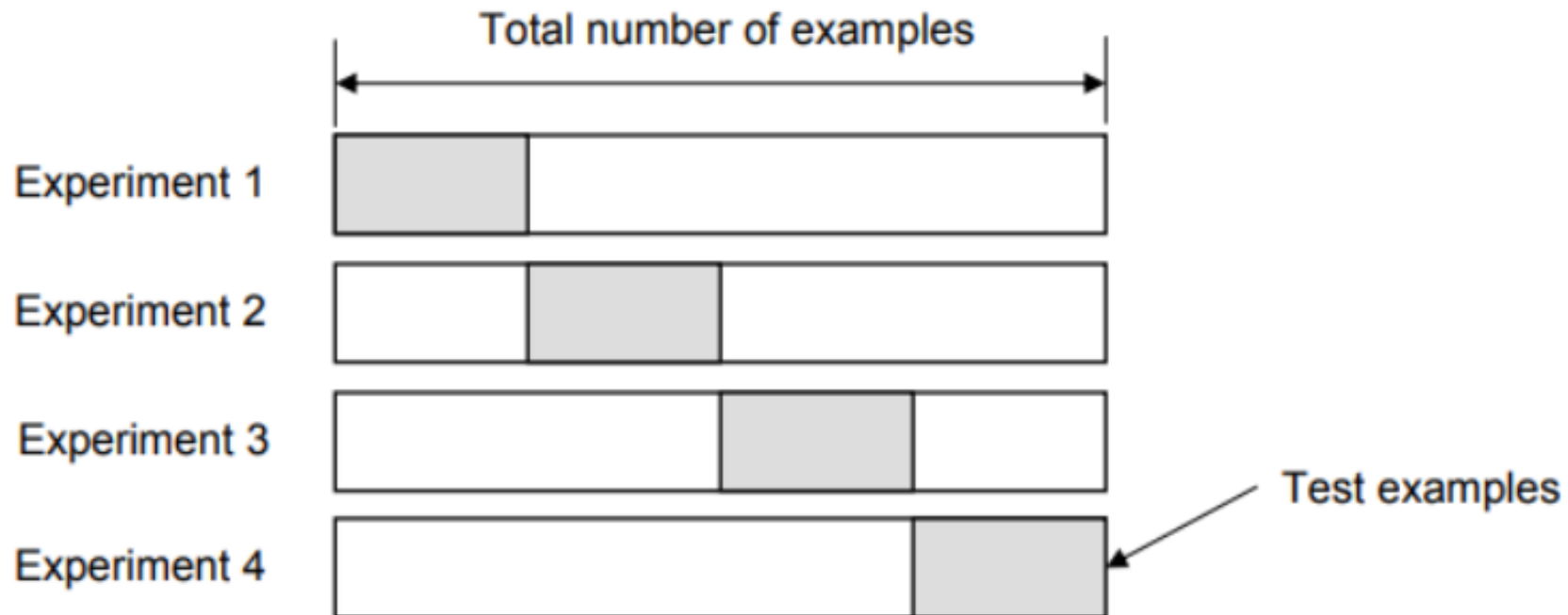


Figure 1: K-fold cross-validation

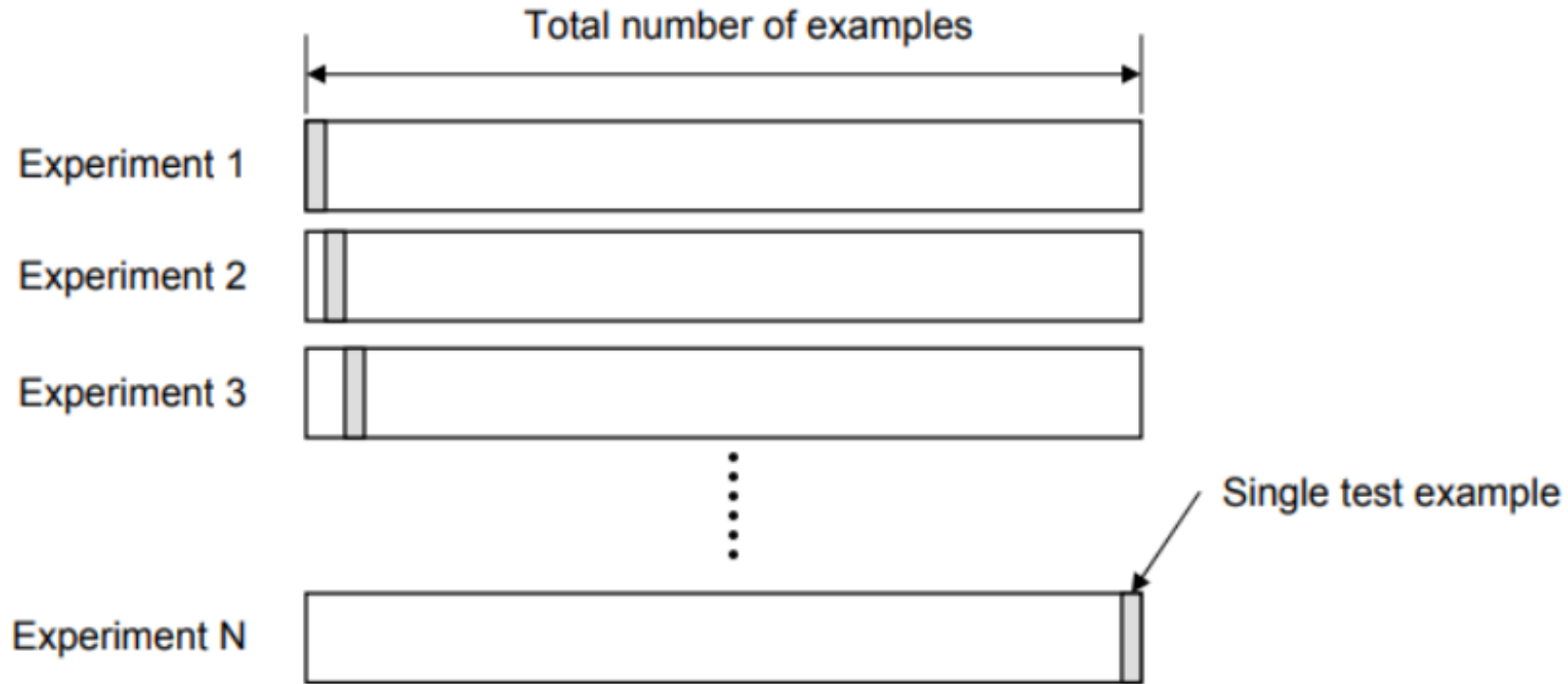
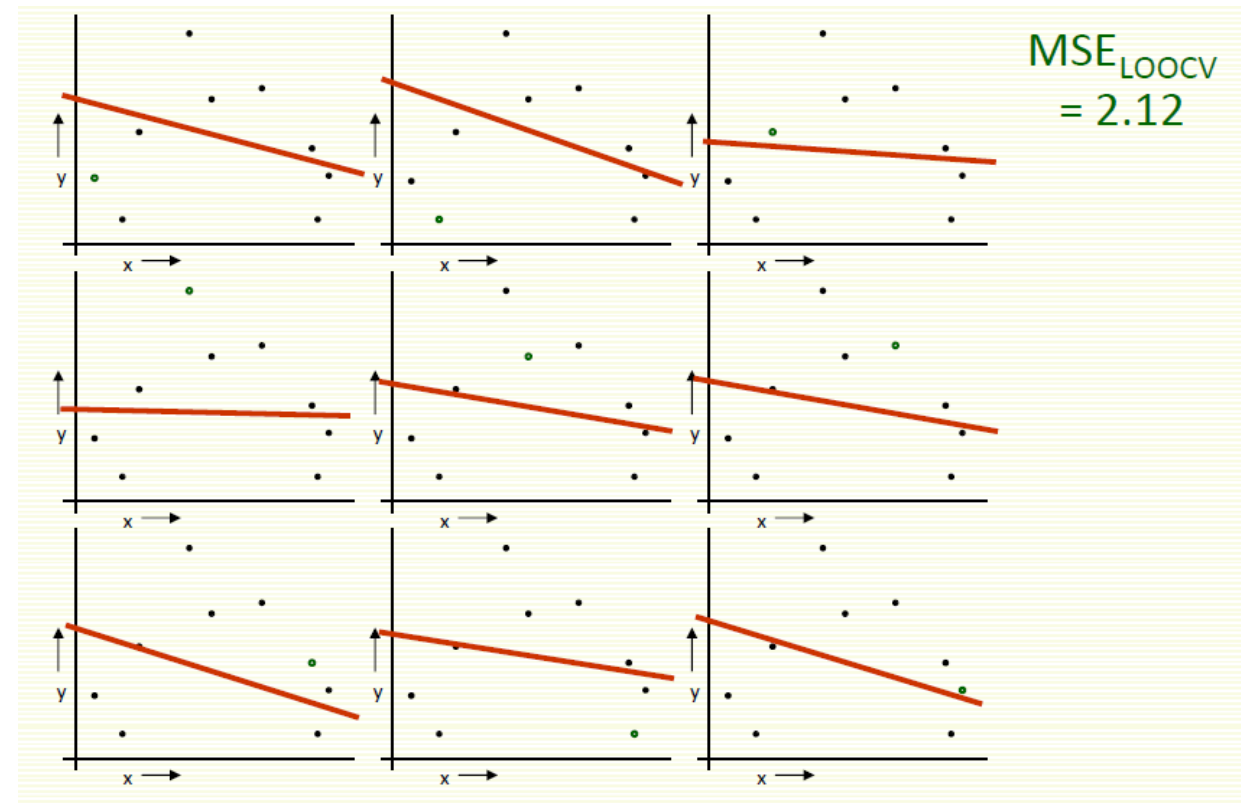
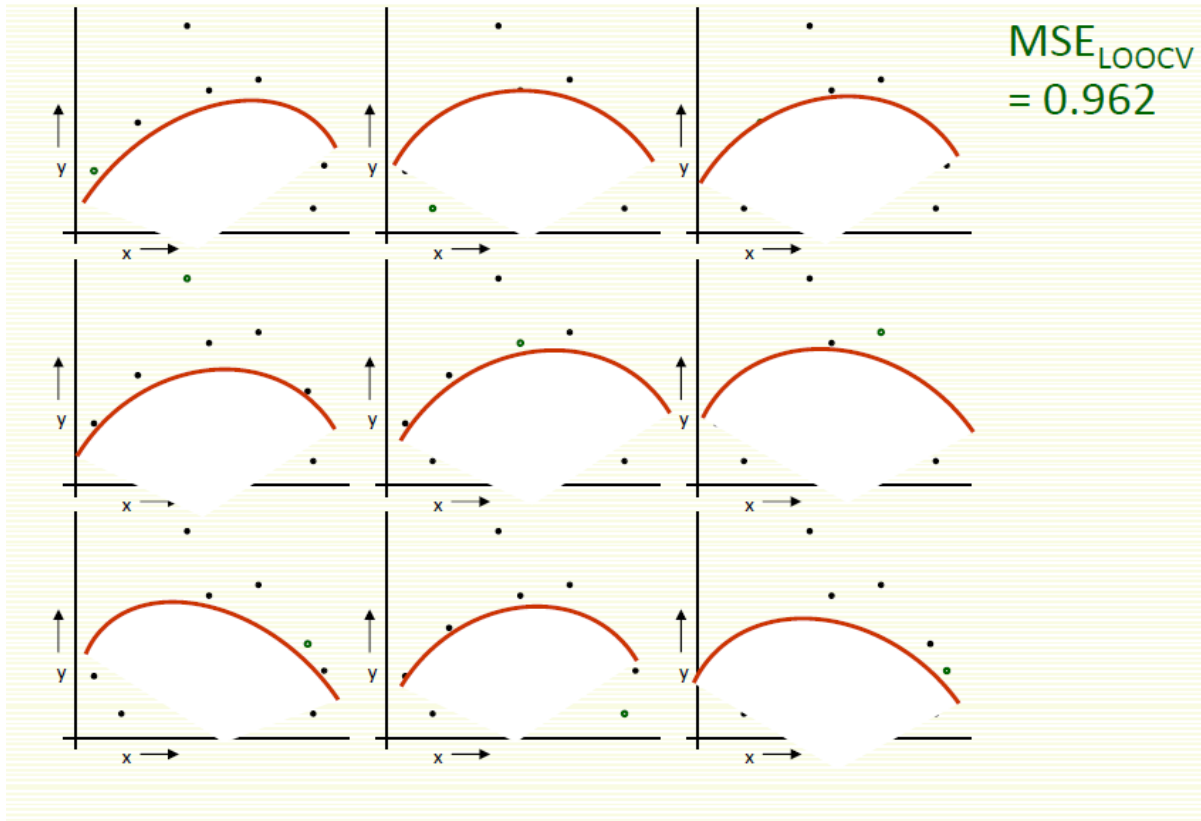


Figure 2: LOOCV validation technique

Leave-One-Out Cross-Validation (LOOCV)

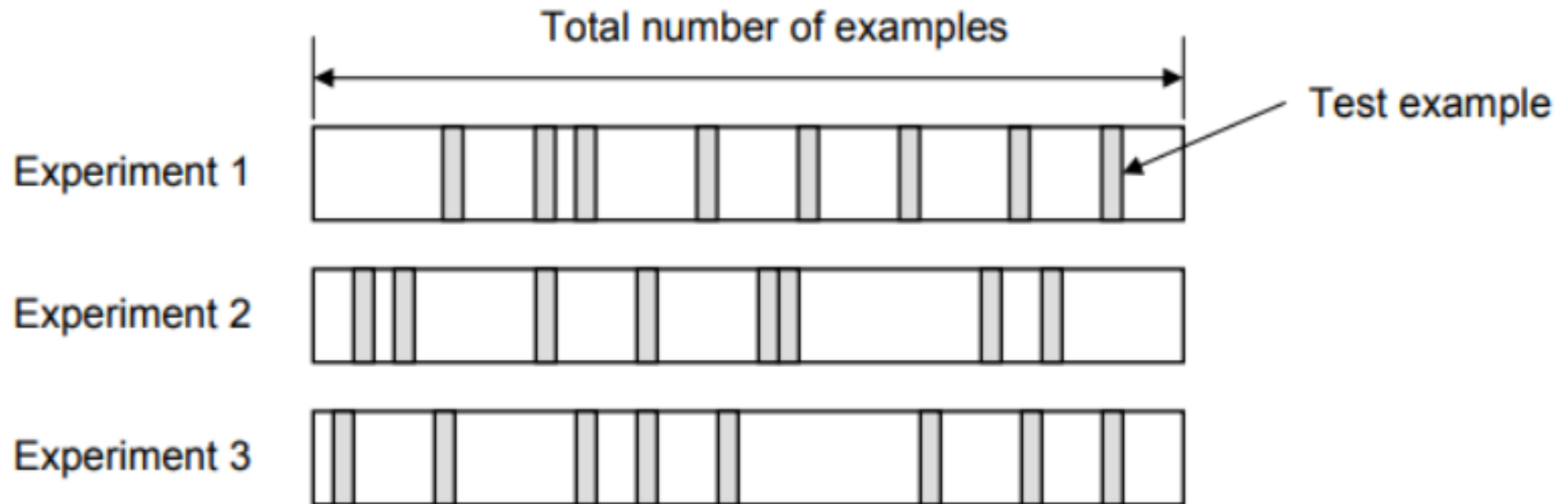
- All the data except one record is used for training, and one record is used for testing
- This process is repeated N times
- `from sklearn.model_selection import LeaveOneOut, LeavePOut`

LOOCV - examples



Random Subsampling

Multiple sets of data are randomly chosen from the dataset and combined to form a test dataset



```
from sklearn.model_selection import ShuffleSplit  
cv = ShuffleSplit(n_splits=7, train_size=0.75, test_size=0.25)
```

Bootstrapping

the training dataset is randomly selected with replacement



What kind of Cross Validation?

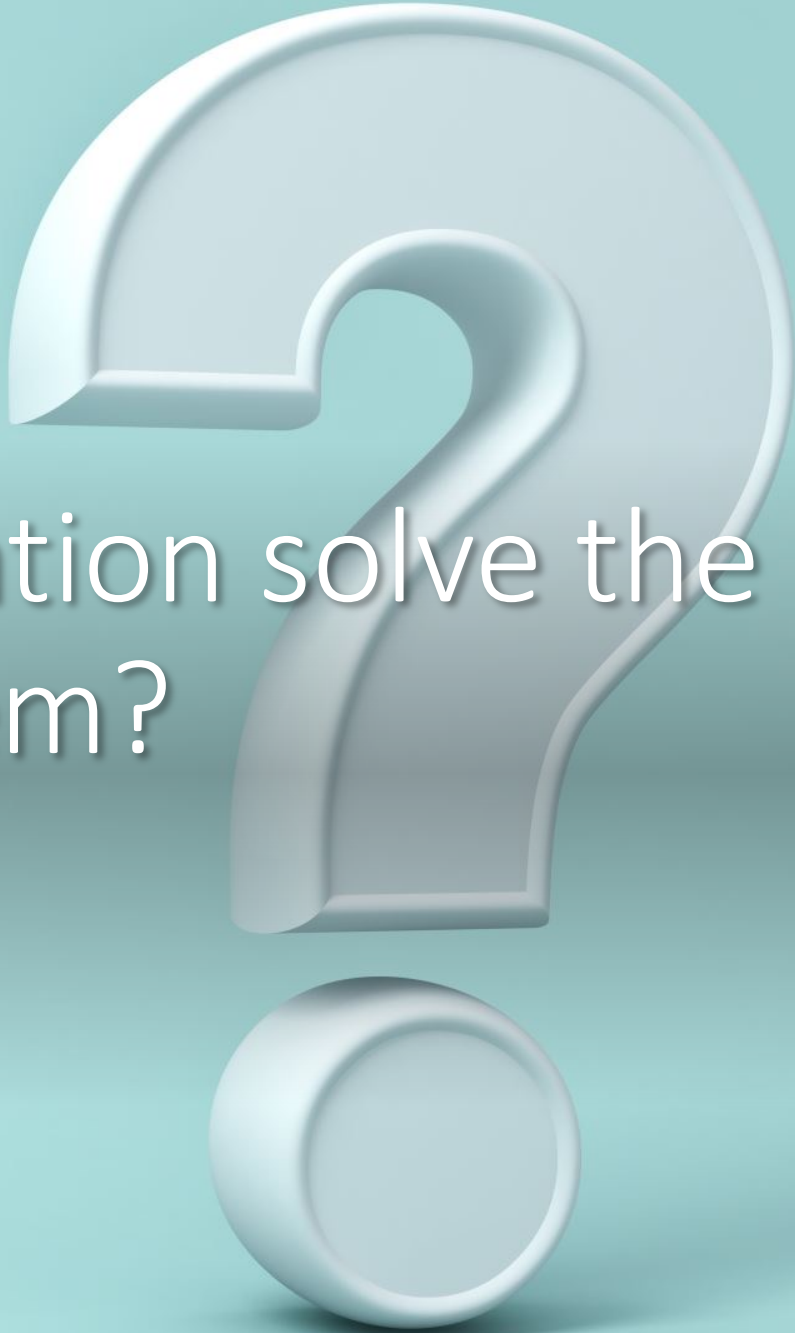
	Downside	Upside
Test-set	may give unreliable estimate of future performance	cheap
Leave-one-out	expensive	doesn't waste data
10-fold	wastes 10% of the data, 10 times more expensive than test set	only wastes 10%, only 10 times more expensive instead of n times
3-fold	wastes more data than 10-fold, more expensive than test set	slightly better than test-set
N-fold	Identical to Leave-one-out	

Not only for tabular data...



Image due to:

<https://towardsdatascience.com/spatial-cross-validation-using-scikit-learn-74cb8ffe0ab9>



Does cross-validation solve the
problem?

Repeated CV

- Repeat the whole process

Pros?

Cons?



```
cv = RepeatedKFold(n_splits=10, n_repeats=3, random_state=1)
# evaluate model
scores = cross_val_score(model, X, y, scoring='accuracy', cv=cv, n_jobs=-1)
# report performance
print('Accuracy: %.3f (%.3f)' % (mean(scores), std(scores)))
```



```
import numpy as np
from sklearn.model_selection
import RepeatedKFold
X = np.array([[1, 2], [3, 4], [1, 2], [3, 4]])
y = np.array([0, 0, 1, 1])
rkf = RepeatedKFold(n_splits=2, n_repeats=2, random_state=2652124)
for train_index, test_index in rkf.split(X):
    print("TRAIN:", train_index, "TEST:", test_index)
    X_train, X_test = X[train_index], X[test_index]
    y_train, y_test = y[train_index], y[test_index]
```

```
TRAIN: [0 1] TEST: [2 3]
TRAIN: [2 3] TEST: [0 1]
TRAIN: [1 2] TEST: [0 3]
TRAIN: [0 3] TEST: [1 2]
```

Validation Data

- Say that you are validating a polynomial regression
- How do you choose the degree?
- **Solution:** split the labeled data into three parts

